Gianluca Peri

Curriculum Vitae

gianluca.peri@unifi.it
www.gianlucaperi.com
gianluca-peri

Education

2024-present **PhD in Machine Learning and Complex Systems' Physics**, *University of Florence*, Italy
Supervisor: Prof. Duccio Fanelli
Co-Supervisor: Prof. Francesco Piazza
In collaboration with: Tecnolink S.r.l.
Research Group: CSDC: Center for the Study of Complex Dynamics

2025-2026 **Long Research Stay**, *University of Namur*, Belgium
Supervisor: Prof. Timoteo Carletti
Project: Hypergraph Neural Networks & Random Walks on Loss Landscapes

2021-2024 **Master's Degree in Theoretical Physics**, *University of Florence*, Italy
Supervisor: Prof. Duccio Fanelli
Thesis: Generalized Spectral Formalism for Neural Networks
Grade: 110/110

2017-2021 **Bachelor's Degree in Physics and Astronomy**, *University of Florence*, Italy
Supervisor: Prof. Simone Landi
Thesis: Fluid Models of Supersonic Winds from Stellar Coronae

2011-2016 **Scientific High School Diploma**, *Liceo Scientifico A. M. Enriques Agnoletti*, Sesto Fiorentino, Italy

Academic Jobs

2025-2026 **Science Communicator**, *OpenLab*, Sesto Fiorentino

2025-2026 **Academic Tutor**, *University of Florence*, Florence

Publications

2026 D. Febbe, D. Fanelli, G. Peri, T. Carletti, *Optimal Navigation on Simplicial Complexes*, Preprint. `https://arxiv.org/abs/2607.29450`

2026 G. Peri, D. Febbe, D. Fanelli, *Spectral Higher-Order Neural Networks Have Sharp Expressivity Bounds*, Preprint. `https://arxiv.org/abs/2607.19042`

2026 G. Peri, T. Carletti, D. Fanelli, D. Febbe, *Spectral Higher-Order Neural Networks*, Under review. `https://arxiv.org/abs/2603.28420`

2026 G. Peri, L. Buffoni, G. Chiti, D. Fanelli, R. Marino, A. Nocentini, P. Panti, *Smart Walkers in Discrete Space*, Published in Chaos, Solitons & Fractals. `https://doi.org/10.1016/j.chaos.2026.118356`

2025 G. Peri, L. Chicchi, D. Fanelli, L. Giambagli, *SPectral ARchiteCture Search for neural network models*, npj Artificial Intelligence. `https://doi.org/10.1038/s44387-025-00039-1`

Conference and Workshop Contributions

09/2026 **Talk**, *SIF 2026*, 112th National Congress of the Italian Physical Society, Florence
Title of the contribution: *Spectral Higher-Order Neural Networks on Time Series*

06/2026 **Talk**, *SIFS*, Italian Society of Statistical Physics, Parma
Title of the contribution: *Spectral Higher-Order Neural Networks*

06/2026 **Poster**, *SPDL*, Statistical Physics of Deep Learning School, Como
Title of the contribution: *Spectral Higher-Order Neural Networks*

05/2026 **Poster**, *PhD Day 2026*, University of Florence, Florence
Title of the contribution: *Spectral Higher-Order Neural Networks*
Award: Poster Prize

04/2026 **Talk**, *NaXys*, NaXys Research Day, Namur
Title of the contribution: *Spectral Higher-Order Neural Networks*

12/2025 **Poster**, *BeNet 2025*, 11th Edition of the Belgian Network Research Meeting, Ghent
Title of the contribution: *Spectral Architecture Search for Neural Networks*

07/2025 **Talk**, *Nordita*, Fluctuations in Self-Interacting and Learning Processes, Stockholm
Title of the contribution: *Training Walkers with Reinforcement Learning*

07/2025 **Poster**, *StatPhys29*, 29th International Conference on Statistical Physics, Florence
Title of the contribution: *Spectral Architecture Search for Neural Networks*

07/2024 **Talk**, *COMPENG 2024*, IEEE Workshop on Complexity in Engineering, Florence
Title of the contribution: *Generalized Spectral Formalism for Neural Networks*

Supervision Experience

2026-present **Bachelor's Degree Thesis - Physics and Astronomy**, *University of Florence*, Francesco Fagone, Configuration Entropy as a Learning Measure

2025-present **Master's Degree Thesis - Physics of Complex Systems**, *University of Florence*, Francesco D'Angelo, Spectral LSTM for Financial Time Series

Languages

Italian	Mothertongue
English	Advanced, level C1
French	Beginner, level A2

Programming Skills

Operating Systems: Linux, Windows

Software and Programming Languages: Python, PyTorch, Matplotlib, Jupyter, Git, LaTeX, bash, Visual Studio Code, Vim, C (basic), C++ (basic)

Other Skills: Supervised and unsupervised learning, reinforcement learning, data analysis, numerical simulations, data mining, scientific computing, high-performance computing, software development, version control, Google Colab, HPC Clusters

References

Prof. Duccio Fanelli: Full Professor - *University of Florence*

Prof. Francesco Piazza: Associate Professor, *University of Florence*

Prof. Timoteo Carletti: Full Professor, *University of Namur*

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